

RECONNAISSANCE & NETWORK ENUMERATION

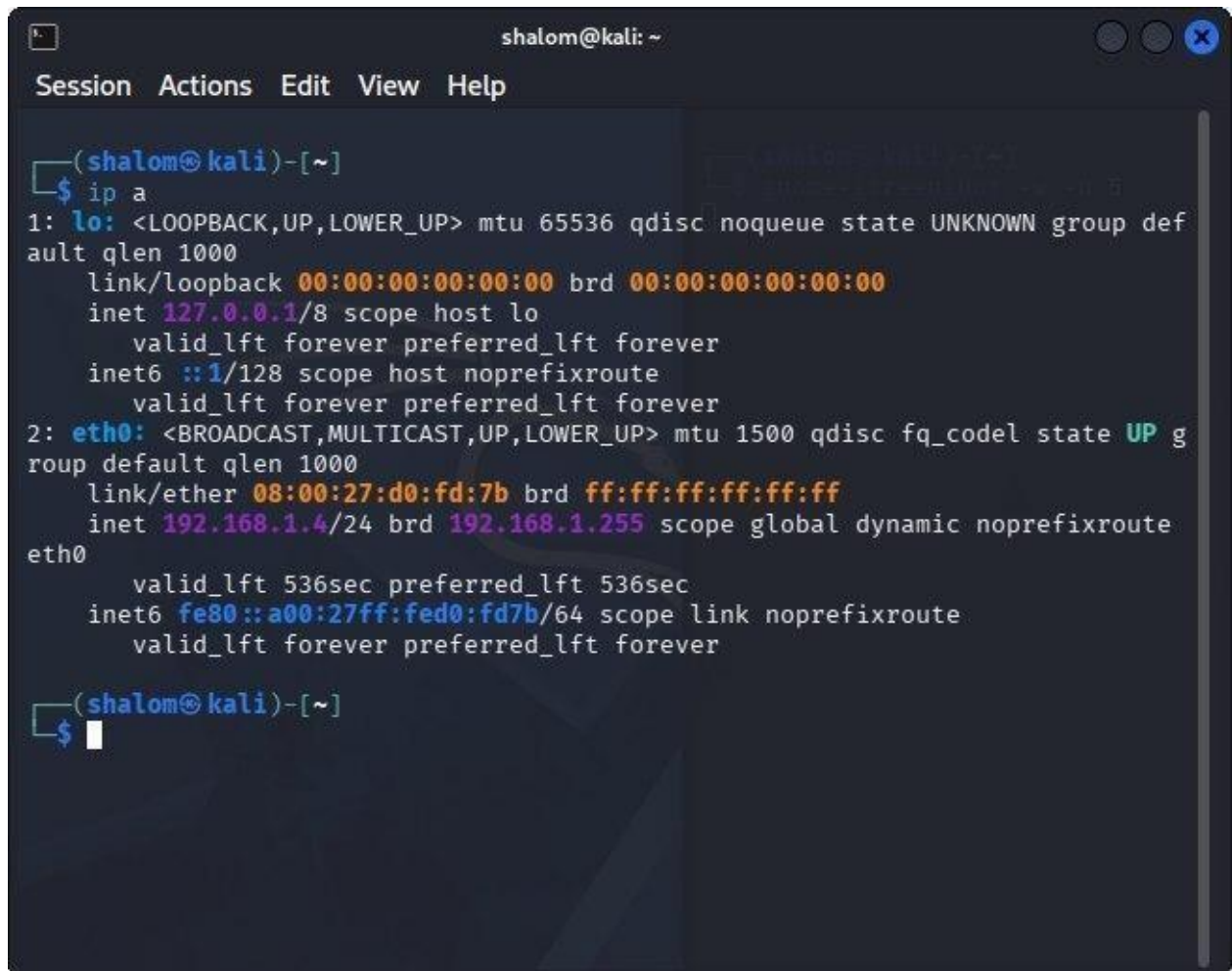
Network Scan Results

Assessment Environment: Authorized VirtualBox laboratory network

Scanner: Kali Linux

Target: Metasploitable 2

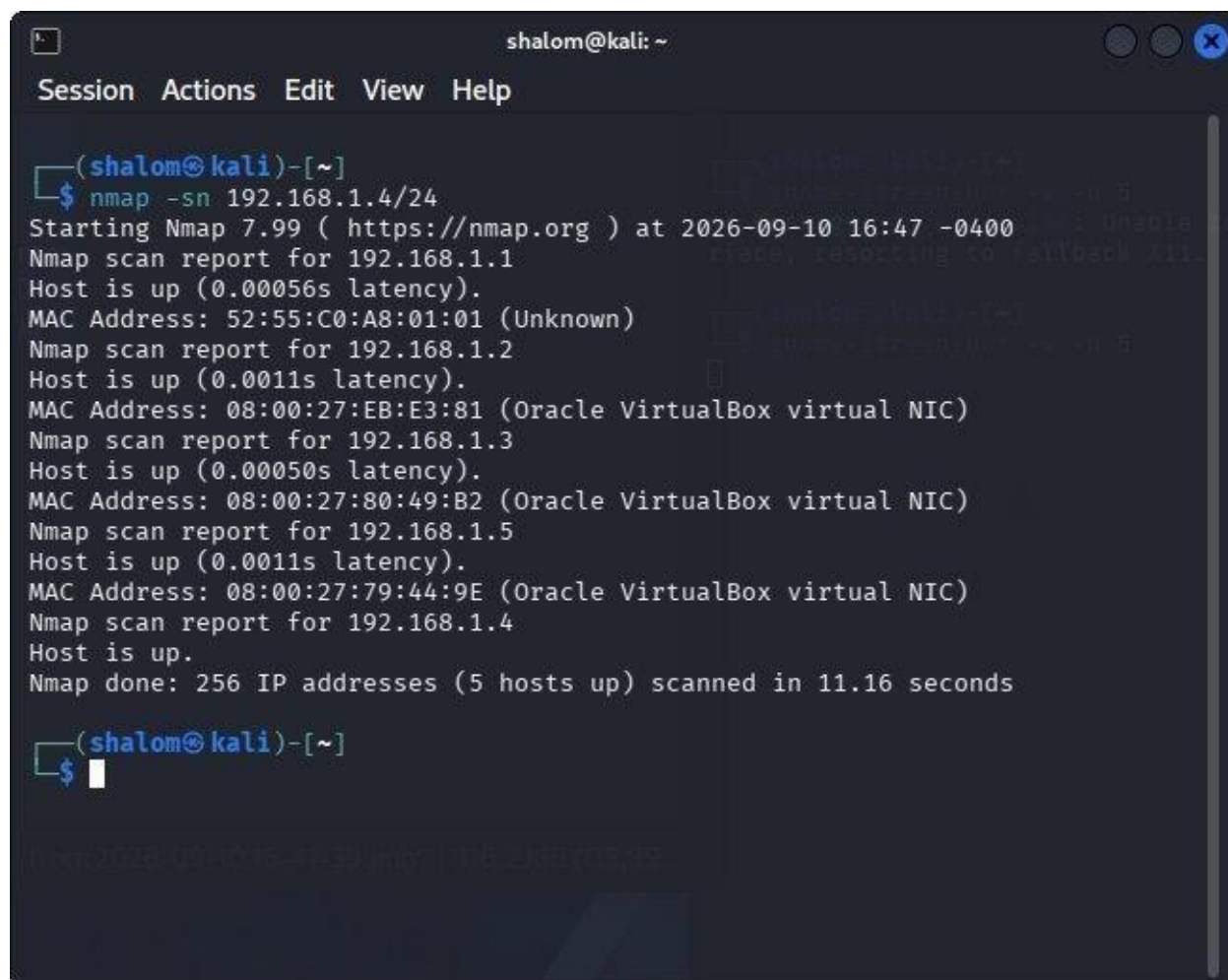
Lab Network Information

A terminal window titled 'shalom@kali: ~' with a menu bar containing 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows the output of the 'ip a' command. It displays details for the loopback interface 'lo' (127.0.0.1) and the ethernet interface 'eth0' (192.168.1.4). The 'eth0' interface is shown as 'UP' and has a MAC address of 08:00:27:d0:fd:7b. The terminal text is as follows:

```
(shalom@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:d0:fd:7b brd ff:ff:ff:ff:ff:ff  
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute  
eth0  
    valid_lft 536sec preferred_lft 536sec  
    inet6 fe80::a00:27ff:fed0:fd7b/64 scope link noprefixroute  
    valid_lft forever preferred_lft forever  
  
(shalom@kali)-[~]  
$
```

Host Discovery

Command executed: `nmap -sn 192.168.1.4/24`

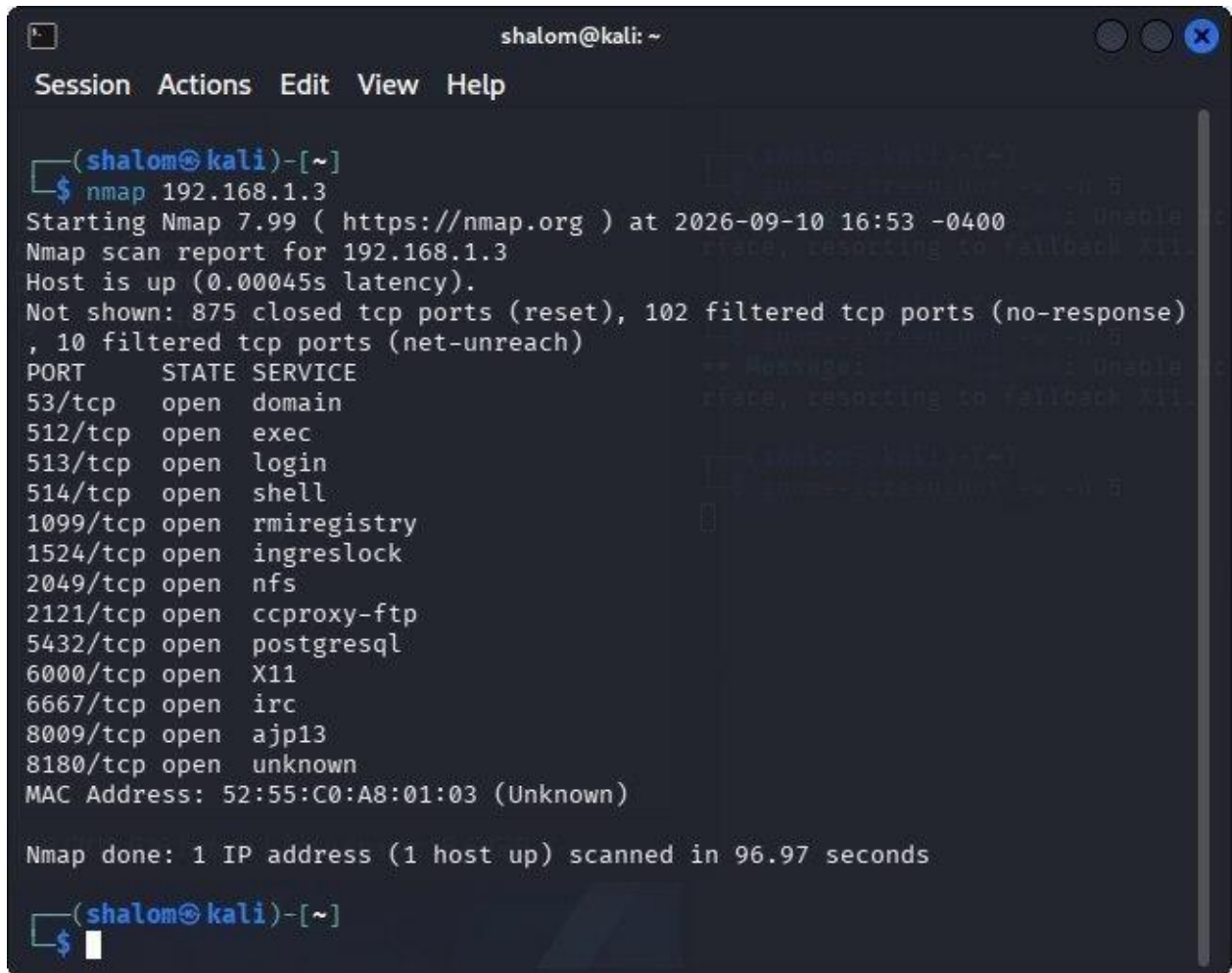
A terminal window titled 'shalom@kali: ~' with a menu bar (Session, Actions, Edit, View, Help). The terminal shows the execution of the command 'nmap -sn 192.168.1.4/24'. The output displays the start of Nmap 7.99, followed by individual scan reports for IP addresses 192.168.1.1, 192.168.1.2, 192.168.1.3, 192.168.1.5, and 192.168.1.4. For 192.168.1.1, the MAC address is 52:55:C0:A8:01:01 (Unknown). For 192.168.1.2, 192.168.1.3, 192.168.1.5, and 192.168.1.4, the MAC addresses are identified as Oracle VirtualBox virtual NICs. The scan concludes by reporting that 256 IP addresses (5 hosts up) were scanned in 11.16 seconds. The prompt returns to '(shalom@kali)-[~]' with a cursor.

```
(shalom@kali)-[~]  
$ nmap -sn 192.168.1.4/24  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-10 16:47 -0400  
Nmap scan report for 192.168.1.1  
Host is up (0.00056s latency).  
MAC Address: 52:55:C0:A8:01:01 (Unknown)  
Nmap scan report for 192.168.1.2  
Host is up (0.0011s latency).  
MAC Address: 08:00:27:EB:E3:81 (Oracle VirtualBox virtual NIC)  
Nmap scan report for 192.168.1.3  
Host is up (0.00050s latency).  
MAC Address: 08:00:27:80:49:B2 (Oracle VirtualBox virtual NIC)  
Nmap scan report for 192.168.1.5  
Host is up (0.0011s latency).  
MAC Address: 08:00:27:79:44:9E (Oracle VirtualBox virtual NIC)  
Nmap scan report for 192.168.1.4  
Host is up.  
Nmap done: 256 IP addresses (5 hosts up) scanned in 11.16 seconds  
  
(shalom@kali)-[~]  
$
```

The host discovery scan reported five live hosts on the /24 network. Three of the discovered addresses were identified as Oracle VirtualBox virtual NICs. The scanner itself was 192.168.1.4, and the system selected for detailed enumeration was 192.168.1.3.

Basic Port Scan

Command executed: nmap 192.168.1.3



```
shalom@kali: ~  
Session Actions Edit View Help  
(shalom@kali)-[~]  
$ nmap 192.168.1.3  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-10 16:53 -0400  
Nmap scan report for 192.168.1.3  
Host is up (0.00045s latency).  
Not shown: 875 closed tcp ports (reset), 102 filtered tcp ports (no-response)  
, 10 filtered tcp ports (net-unreach)  
PORT      STATE SERVICE  
53/tcp    open  domain  
512/tcp    open  exec  
513/tcp    open  login  
514/tcp    open  shell  
1099/tcp   open  rmiregistry  
1524/tcp   open  ingreslock  
2049/tcp   open  nfs  
2121/tcp   open  ccproxy-ftp  
5432/tcp   open  postgresql  
6000/tcp   open  X11  
6667/tcp   open  irc  
8009/tcp   open  ajp13  
8180/tcp   open  unknown  
MAC Address: 52:55:C0:A8:01:03 (Unknown)  
  
Nmap done: 1 IP address (1 host up) scanned in 96.97 seconds  
(shalom@kali)-[~]  
$
```

The basic scan identified the target as reachable and reported 13 open TCP ports. Nmap also reported 875 closed TCP ports that responded with resets.

Service and Version Detection

Command executed: nmap -sV 192.168.1.3

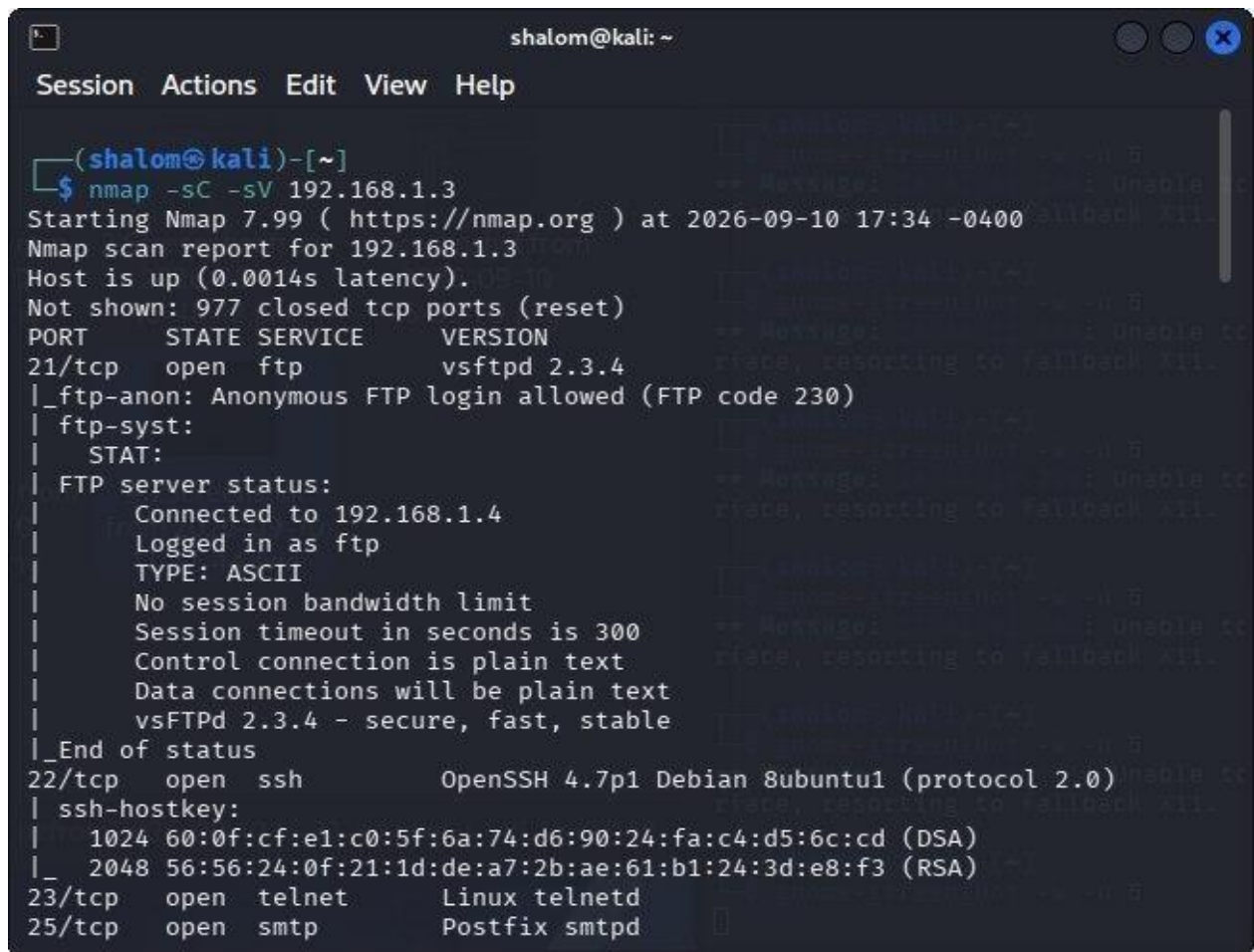
```
shalom@kali: ~  
Session Actions Edit View Help  
(shalom@kali)-[~]  
$ nmap -sV 192.168.1.3  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-10 17:31 -0400  
Nmap scan report for 192.168.1.3  
Host is up (0.0021s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp           vsftpd 2.3.4  
22/tcp    open  ssh           OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet        Linux telnetd  
25/tcp    open  smtp          Postfix smtpd  
53/tcp    open  domain        ISC BIND 9.4.2  
80/tcp    open  http          Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind       2 (RPC #100000)  
139/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec          netkit-rsh rshd  
513/tcp   open  login         OpenBSD or Solaris rlogind  
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi      GNU Classpath grmiregistry  
1524/tcp  open  bindshell     Metasploitable root shell  
2049/tcp  open  nfs           2-4 (RPC #100003)  
2121/tcp  open  ftp           ProFTPD 1.3.1  
3306/tcp  open  mysql         MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql    PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc           VNC (protocol 3.3)  
6000/tcp  open  X11           (access denied)  
6667/tcp  open  irc           UnrealIRCd
```

```
shalom@kali: ~  
Session Actions Edit View Help  
80/tcp    open  http          Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind       2 (RPC #100000)  
139/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec          netkit-rsh rshd  
513/tcp   open  login         OpenBSD or Solaris rlogind  
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi      GNU Classpath grmiregistry  
1524/tcp  open  bindshell     Metasploitable root shell  
2049/tcp  open  nfs           2-4 (RPC #100003)  
2121/tcp  open  ftp           ProFTPD 1.3.1  
3306/tcp  open  mysql         MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql    PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc           VNC (protocol 3.3)  
6000/tcp  open  X11           (access denied)  
6667/tcp  open  irc           UnrealIRCd  
8009/tcp  open  ajp13         Apache Jserv (Protocol v1.3)  
8180/tcp  open  http          Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 52:55:C0:A8:01:03 (Unknown)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs : Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 18.36 seconds  
(shalom@kali)-[~]  
$
```

Service/version detection provided additional information about the software exposed on the open ports.

Script and Host Enumeration Findings

Command executed: `nmap -sC -sV 192.168.1.3`



```
shalom@kali: ~  
Session Actions Edit View Help  
  
(shalom@kali)-[~]  
$ nmap -sC -sV 192.168.1.3  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-10 17:34 -0400  
Nmap scan report for 192.168.1.3  
Host is up (0.0014s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)  
|_ftp-syst:  
|   STAT:  
|_FTP server status:  
|   Connected to 192.168.1.4  
|   Logged in as ftp  
|   TYPE: ASCII  
|   No session bandwidth limit  
|   Session timeout in seconds is 300  
|   Control connection is plain text  
|   Data connections will be plain text  
|   vsFTPD 2.3.4 - secure, fast, stable  
|_End of status  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
|_ssh-hostkey:  
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)  
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd
```

```
shalom@kali: ~  
Session Actions Edit View Help  
|_ssl-date: 2026-09-10T21:34:54+00:00; 0s from scanner time.  
|_ssl-v2:  
|   SSLv2 supported  
|   ciphers:  
|     SSL2_RC2_128_CBC_WITH_MD5  
|     SSL2_RC4_128_WITH_MD5  
|     SSL2_DES_192_EDE3_CBC_WITH_MD5  
|     SSL2_DES_64_CBC_WITH_MD5  
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5  
|     SSL2_RC4_128_EXPORT40_WITH_MD5  
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=0  
COSA/stateOrProvinceName=There is no such thing outside US/countryName=XX  
|_Not valid before: 2010-03-17T14:07:45  
|_Not valid after: 2010-04-16T14:07:45  
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY,  
ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN  
53/tcp open domain ISC BIND 9.4.2  
|_dns-nsid:  
|_bind.version: 9.4.2  
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
|_http-title: Metasploitable2 - Linux  
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2  
111/tcp open rpcbind 2 (RPC #100000)  
|_rpcinfo:  
|   program version port/proto service  
|   100005 1,2,3 51430/udp mountd  
|   100005 1,2,3 59653/tcp mountd  
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
```

```
shalom@kali: ~  
Session Actions Edit View Help  
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)  
512/tcp open exec netkit-rsh rexecd  
513/tcp open login OpenBSD or Solaris rlogind  
514/tcp open tcpwrapped  
1099/tcp open java-rmi GNU Classpath grmiregistry  
1524/tcp open bindshell Metasploitable root shell  
2049/tcp open nfs 2-4 (RPC #100003)  
2121/tcp open ftp ProFTPD 1.3.1  
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5  
|_mysql-info:  
|   Protocol: 10  
|   Version: 5.0.51a-3ubuntu5  
|   Thread ID: 9  
|   Capabilities flags: 43564  
|   Some Capabilities: SwitchToSSLAfterHandshake, Support41Auth, Speaks41Prot  
ocolNew, LongColumnFlag, SupportsTransactions, SupportsCompression, ConnectWi  
thDatabase  
|   Status: Autocommit  
|   Salt: G.N~Sbs<o<{ym7g%}hr'  
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7  
|_ssl-date: 2026-09-10T21:34:54+00:00; 0s from scanner time.  
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=0  
COSA/stateOrProvinceName=There is no such thing outside US/countryName=XX  
|_Not valid before: 2010-03-17T14:07:45  
|_Not valid after: 2010-04-16T14:07:45  
5900/tcp open vnc VNC (protocol 3.3)  
|_vnc-info:  
|   Protocol version: 3.3
```

```
shalom@kali: ~  
Session Actions Edit View Help  
| Security types:  
|_ VNC Authentication (2)  
6000/tcp open X11 (access denied)  
6667/tcp open irc UnrealIRCd  
| irc-info: 09:30 2026-09-10  
| users: 1  
| servers: 1  
| lusers: 1  
| lservers: 0  
| server: irc.Metasploitable.LAN  
| version: Unreal3.2.8.1. irc.Metasploitable.LAN  
| uptime: 0 days, 0:18:58  
| source ident: nmap  
| source host: 5765EFD5.78DED367.FFFA6D49.IP  
|_ error: Closing Link: ddvlstzqw[192.168.1.4] (Quit: ddvlstzqw)  
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)  
|_ajp-methods: Failed to get a valid response for the OPTION request  
8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1  
|_http-favicon: Apache Tomcat  
|_http-title: Apache Tomcat/5.5  
|_http-server-header: Apache-Coyote/1.1  
MAC Address: 52:55:C0:A8:01:03 (Unknown)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs  
: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Host script results:  
| smb-os-discovery:  
| OS: Unix (Samba 3.0.20-Debian)
```

```
shalom@kali: ~  
Session Actions Edit View Help  
MAC Address: 52:55:C0:A8:01:03 (Unknown)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs  
: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Host script results:  
| smb-os-discovery:  
| OS: Unix (Samba 3.0.20-Debian)  
| Computer name: metasploitable  
| NetBIOS computer name:  
| Domain name: localdomain  
| FQDN: metasploitable.localdomain  
|_ System time: 2026-09-10T17:34:46-04:00  
|_clock-skew: mean: 1h00m00s, deviation: 2h00m01s, median: 0s  
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC:  
<unknown> (unknown)  
|_smb2-time: Protocol negotiation failed (SMB2)  
| smb-security-mode:  
| account_used: guest  
| authentication_level: user  
| challenge_response: supported  
|_ message_signing: disabled (dangerous, but default)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 27.83 seconds  
  
(shalom@kali)-[~]  
$
```

The default-script and version scan supplied additional reconnaissance information. Important observations included the following:

FTP (21/tcp): Nmap reported that anonymous FTP login was allowed. The FTP service also reported that control and data connections were plain text.

SMTP (25/tcp): The scan reported SSLv2 support and identified SMTP capabilities including VRFY and STARTTLS. The certificate information shown in the output was dated 2010.

HTTP (80/tcp): The web service identified itself as Apache 2.2.8 on Ubuntu and the page title was 'Metasploitable2 - Linux'.

Key Reconnaissance Results

Finding	Result
Live hosts discovered	5
Detailed target	192.168.1.3
Open TCP ports identified	13
Closed TCP ports reported	875
Target operatingsystem family	Unix/Linux (Nmap service information)
Notable exposed technologies	FTP, SSH, Telnet, SMTP, DNS, HTTP, SMB, NFS, MySQL, PostgreSQL, VNC, IRC, Java

Conclusion

The network reconnaissance successfully identified live hosts within the laboratory network and enumerated the open TCP ports and services on the selected Metasploitable 2 target. Nmap version detection and default scripts provided additional information about software versions, host details, and selected service configurations. These results establish the technical evidence needed for the next deliverable, Service Enumeration Report, and provide a basis for summarizing the target's attack surface.